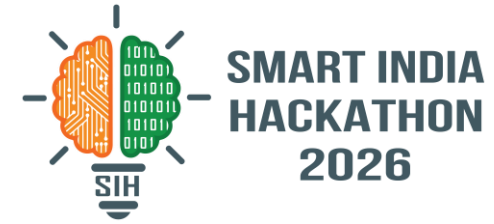
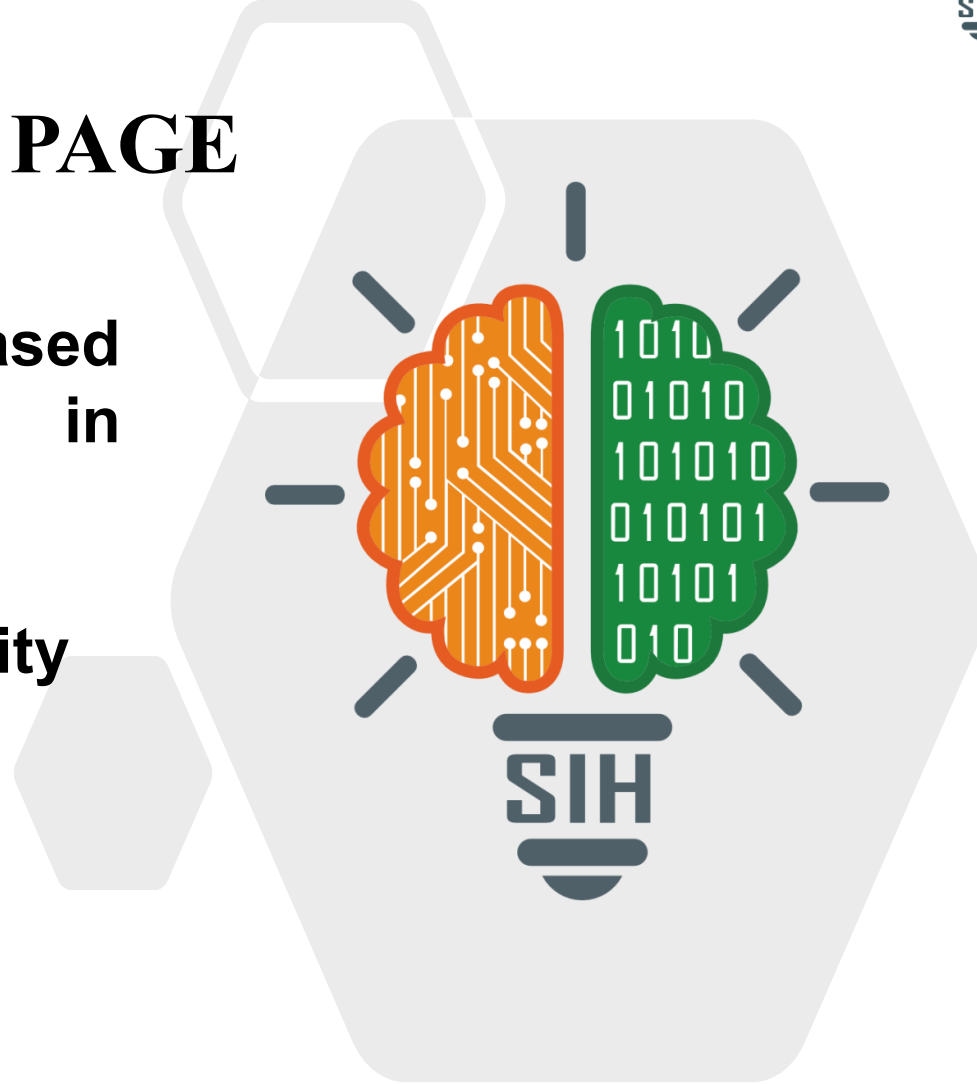


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TITLE PAGE

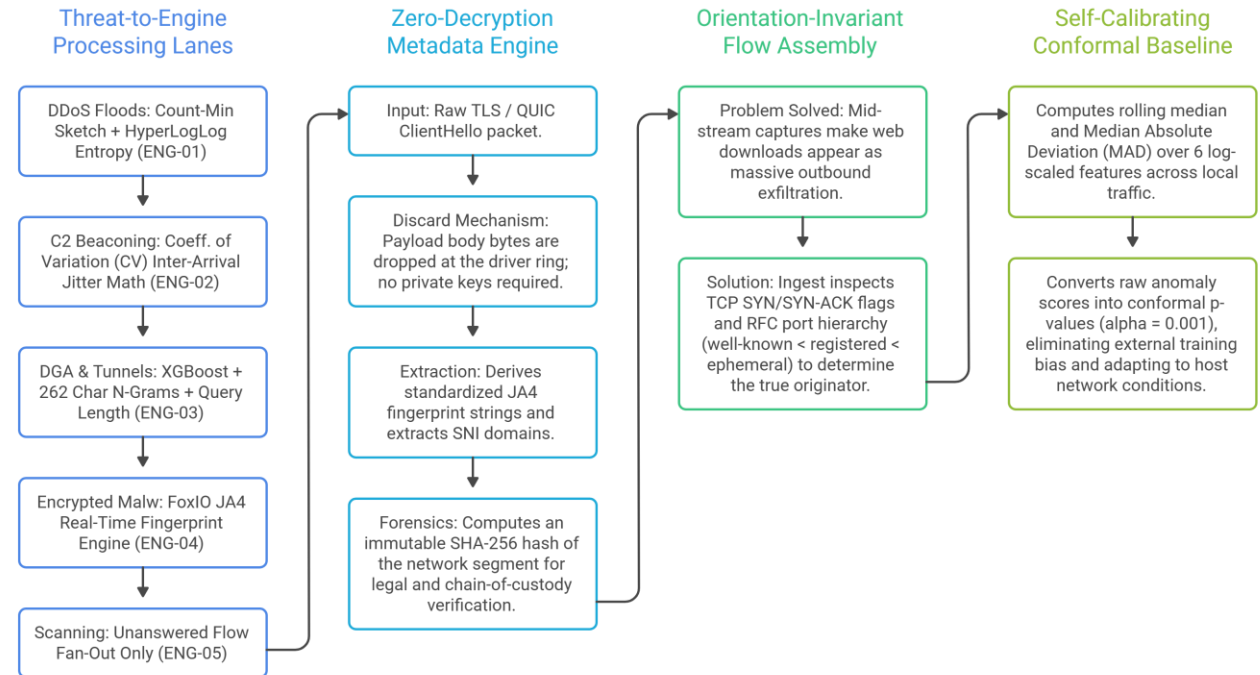
- Problem Statement ID – 26145
- Problem Statement Title- AI-Based Detection of Cyber Threats in Unidirectional IP Traffic
- Theme- Blockchain & Cybersecurity
- PS Category- Software
- Team ID- 135060
- Team Name : TeamXOR

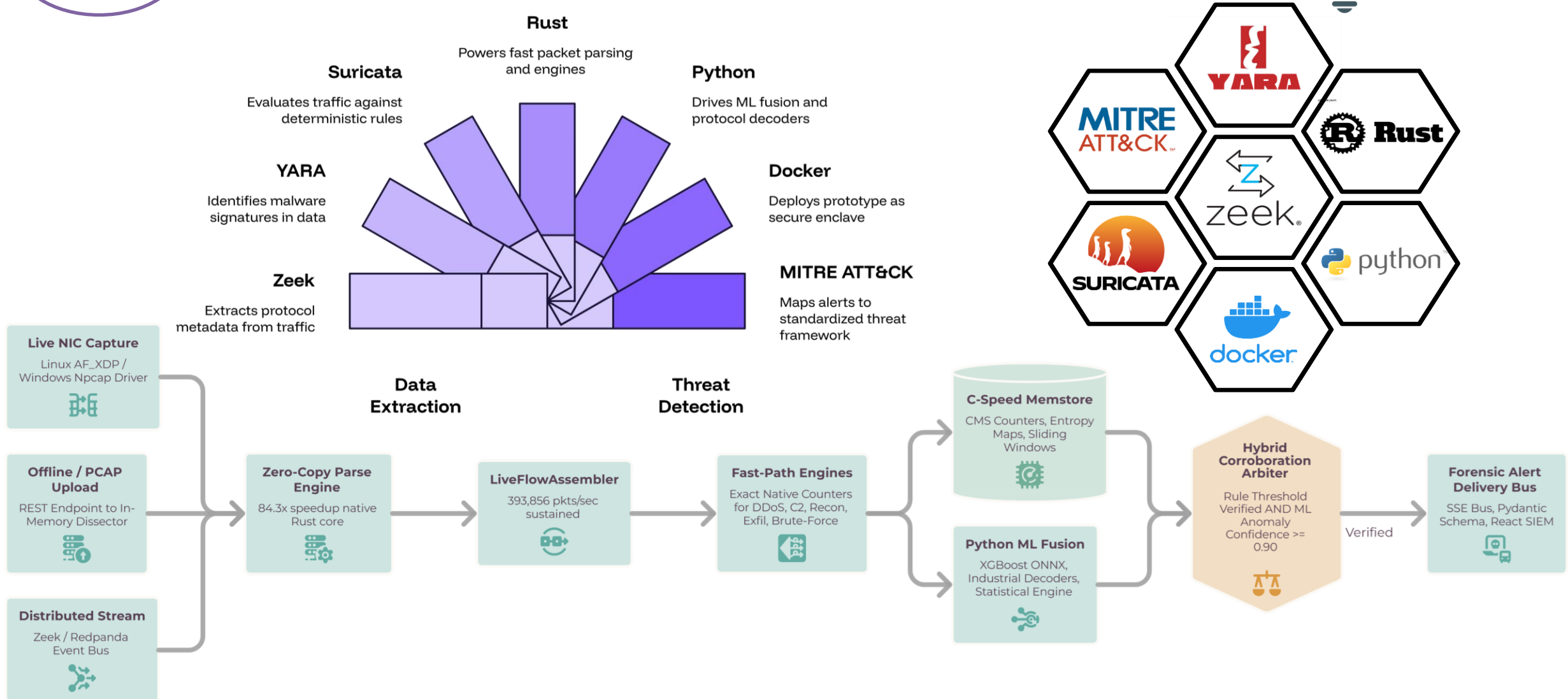


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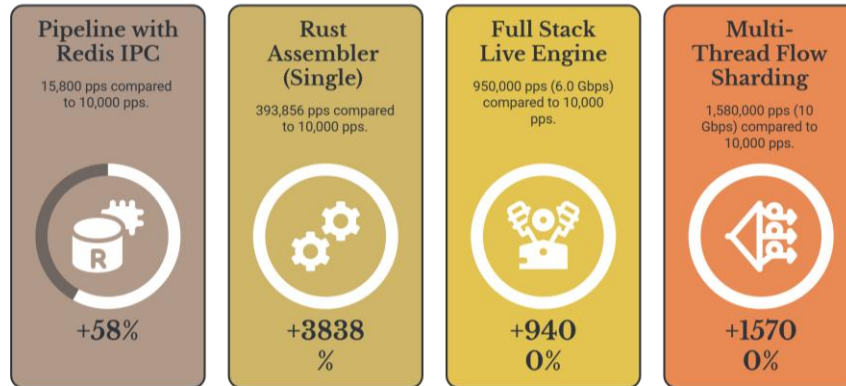
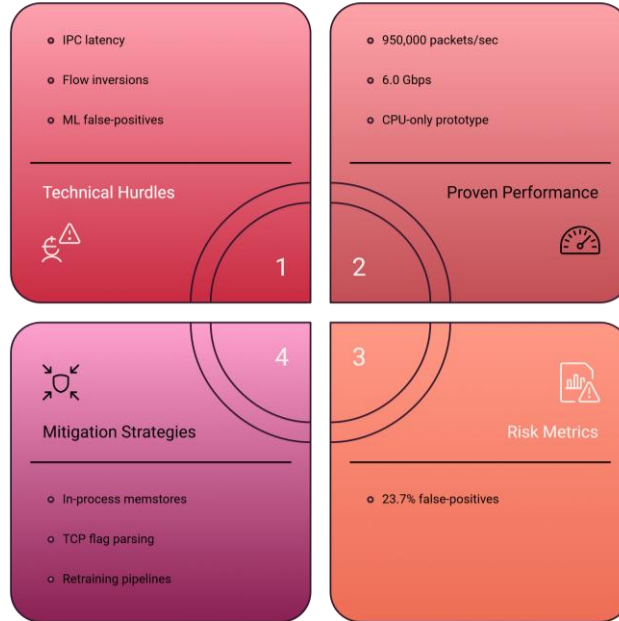
- **Detailed explanation of the proposed solution:** A secure, read-only AI pipeline that detects and scores threats purely from network metadata without ever decrypting payloads.
- **How it addresses the problem:** It processes four different data ingest paths through a single, near real-time detection stack outputting to a MITRE-mapped dashboard.
- **Innovation and uniqueness of the solution:** Uses specialized mechanisms per threat, hybrid ML to prevent train/serve skew, and JA4 TLS fingerprinting for zero-decryption analysis.

StealthTap Detection Engines and Component Workflows





FEASIBILITY AND VIABILITY



Rust-based solutions significantly outperform Python and Redis IPC, with multi-threaded flow sharding achieving the highest throughput.

Mid-Stream
Inversion

IPC Redis
Bottleneck

Model In-Sample
Skew

Local Discovery
Noise

Empirical Impact Observed	Engineering Mitigation Applied	Measured Production Result
118 KB downloads flagged as 86:1 outbound exfiltration.	Enforced TCP SYN/SYN-ACK parsing and port rank precedence.	Zero false exfiltration alerts on mid-stream sessions.
Localhost socket I/O capped processing at 1,580 pps.	Ported engines to native Rust using in-process MemoryStore.	Throughput scaled to ~950,000 pps (6.0 Gbit/s sustained).
Pre-trained DDoS model had 23.7% false positive rate on live links.	Retrained on real network baselines + enforced rule corroboration.	False positive rate dropped to 0.00% at >= 0.9 threshold.
SSDP / mDNS broadcasts triggered anomaly alerts.	Excluded discovery-multicast ranges from baseline training.	0 false alerts across 18,053 discovery frames.

IMPACT AND BENEFITS

Target Audience Impact

Closes a critical blind spot for NTR, SOCs, and infrastructure operators by enabling near real-time, fully auditable detection on strictly one-way links.

Security & Operational Benefits

Provides comprehensive protection across IT and OT/ICS threats including Modbus, DNP3, and EtherNet/IP from a single passive tap.

Economic Benefits

Utilizes an open-source architecture on commodity CPU hardware, eliminating expensive proprietary per-sensor licensing costs.



Impact and Benefits

Economic Scalability
60%

6.0 Gbit/s threat detection

High-Resilience Autonomous Deployment
70%

Maximum uptime guaranteed

Deep OT/ICS Infrastructure Protection
80%

Protects critical infrastructure

Evidentiary Forensic Integrity
90%

Clean chain of custody

Absolute Network Isolation
100%

Zero attack surface



Current Situation



Fails to inspect modern encrypted TLS/QUIC sessions

Lacks decryption capabilities without private keys



Suffers from significant train-versus-serve skew

Accuracy drops to 3–14% on industrial protocols



Lacks deep inspection for OT/ICS environments

Blind to localized sabotage and unauthorized commands



Prone to mid-stream inversion false alarms

Misidentifies normal downloads as data exfiltration



StealthTap Impact



Extracts FoxIO JA4 metadata and SNI

Bypasses payload decryption requirements entirely



Utilizes self-calibrating conformal baselines

Achieves near-zero false positive rates



Decodes 7 critical industrial protocols

Enables real-time mapping of control commands



Employs orientation-invariant flow assembly

Relies on TCP flags and RFC port ranks

Detection engines & tools

Zeek (network security monitor) — <https://github.com/zeek/zeek>
Suricata (IDS engine) — <https://github.com/OISF/suricata>
Suricata ET Open ruleset — <https://rules.emergingthreats.net/>
YARA (malware/file pattern matching) — <https://github.com/VirusTotal/yara>
ICSNPP (Zeek OT/ICS protocol parsers) — <https://github.com/cisagov/ICSNPP>
RITA (beacon-hunting methodology reference) — <https://github.com/activecm/rita>

ML models & standards

FoxIO JA4 TLS-fingerprinting spec — <https://github.com/FoxIO-LLC/ja4>
Chen & Guestrin, "XGBoost: A Scalable Tree Boosting System" (2016) — <https://arxiv.org/abs/1603.02754>
Liu, Ting & Zhou, "Isolation Forest" (ICDM 2008) —
<https://cs.nju.edu.cn/zhoush/zhoush.files/publication/icdm08b.pdf>
ONNX Runtime — <https://github.com/onnx/onnx>
RFC 6238 (TOTP, multi-factor auth) — <https://www.rfc-editor.org/rfc/rfc6238>

Threat-model references

MITRE ATT&CK for ICS — <https://attack.mitre.org/matrices/ics/>
CVE-2004-2687 (distcc RCE) — <https://www.cve.org/CVERecord?id=CVE-2004-2687>

Training & validation datasets

DGA domain dataset (dns model, 675k domains) — https://github.com/chrnor/DGA_domains_dataset
Cucchiarelli et al., DGA detection paper, *Expert Systems with Applications* —
<https://doi.org/10.1016/j.eswa.2020.114551>
Attack-scenario PCAP corpus (25 real attack captures) — <https://www.kaggle.com/datasets/olaidegabriel/attack-scenario-dataset-in-pcap-format>
ITI/ICS-Security-Tools (Modbus/CIP/S7 test captures) — <https://github.com/ITI/ICS-Security-Tools>
Digital Bond Quickdraw-Snort (ICS protocol test captures) — <https://github.com/digitalbond/Quickdraw-Snort>
Yara-Rules (community YARA signatures) — <https://github.com/Yara-Rules/rules>
Awesome-Cybersecurity-Datasets (index) — <https://github.com/shramos/Awesome-Cybersecurity-Datasets>

Project Github Repo
[Click Here](#) or Scan Below QR



Reference: <https://github.com/SARDHAK-CYBER/SIH26145-SIH2026.git>